Who Votes in Chicago? Participation and Representativeness in Presidential Elections, 2008-2024

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Motivation and Research Question

Democratic systems rely on broad participation in elections, yet voter turnout rarely occurs evenly across communities. In cities like Chicago, participation varies widely across neighborhoods with different socioeconomic conditions. These differences matter because turnout ultimately shapes whose preferences are represented in democratic decision-making. When some communities consistently participate at higher rates than others, the electorate may not fully reflect the population as a whole.

This project examines who votes in Chicago and how representative the electorate is across neighborhoods during presidential elections from 2008 through 2024. Specifically, we ask two related questions. First, how does voter turnout vary spatially across Chicago precincts? Second, how strongly are turnout patterns associated with neighborhood socioeconomic and demographic characteristics?

Understanding these patterns has practical implications for policymakers, civic organizations, and election administrators. If participation gaps are concentrated in particular types of neighborhoods, targeted policies may help reduce barriers to participation and strengthen democratic representation. By analyzing turnout across five presidential elections that capture varying political contexts, this project provides a long-term perspective on how participation patterns persist across time and how they relate to structural neighborhood characteristics.

Our goal is not to establish causal relationships but to provide a clear descriptive policy-oriented evidence of turnout inequality across Chicago and explore what these patterns imply for civic engagement and public policy.

Data and Methodology

To analyze voting behavior across the city, we combine precinct-level election results with demographic and spatial data from several sources.

First, we use voter turnout data from the Chicago Board of Elections, which provide counts of registered voters and ballots cast for each precinct during presidential elections from 2008 to 2024. These data allow us to calculate turnout rates for thousands of precinct observations across five election cycles.

Second, we incorporate neighborhood demographic characteristics from the American Community Survey (ACS). We use ACS 5-year estimates aligned to each election cycle to approximate neighborhood conditions during that period. The demographic variables we focus on include median household income, college education, renter share, racial composition, and the share of young residents (age 18–29), all measured at the census tract level. These variables capture important socioeconomic conditions that may influence voter participation. To ensure the analysis focuses strictly on the city, ACS census tracts are clipped to the Chicago municipal boundary, removing areas outside city limits.

Third, we use geographic boundary files for both census tracts and voting precincts. Precinct boundary files are obtained from the Chicago Data Portal, while census tract shapefiles come from the U.S. Census TIGER/Line program. Because these geographic units do not share identical boundaries, demographic information cannot be merged directly onto precinct-level election data. To address this challenge, we apply area-weighted spatial interpolation, which calculates the geographic overlap between census tracts and precinct polygons and assigns demographic characteristics to precincts proportionally based on their overlapping area.

The resulting dataset is a **precinct-by-year panel**, where each observation represents a precinct in a particular election year and includes turnout rates, demographic characteristics, and geographic identifiers.

To summarize neighborhood conditions more clearly, we construct a **Socioeconomic Status (SES) index** using Principal Component Analysis (PCA). This index combines several demographic characteristics into a single measure capturing overall neighborhood socioeconomic advantage.

Together, the final master data allow us to examine how turnout varies spatially across Chicago and how strongly participation is associated with neighborhood characteristics.

Part 8 – Socioeconomic Status Index (SES)

Analysis Approach and Dashboard

Our project combines static visual analysis with an interactive dashboard that allows users to explore spatial turnout patterns across the City, relationship between turnout and socioeconomic status, and how demographic correlations with turnout evolve across election cycles.

Findings

Spatial patterns in voter turnout

Our first visualization maps turnout rates across Chicago precincts for each presidential election.

The maps reveal clear and persistent spatial patterns in voter participation. Higher-turnout precincts consistently appear in parts of the North Side and lakefront neighborhoods, while lower turnout is more common in parts of the South and West sides of the city. Importantly, these spatial patterns remain visible across all five election years, illustrating that turnout differences are geographically clustered rather than randomly distributed across the city.

Even in high-turnout presidential elections (2008 and 2012), relative differences between neighborhoods remain strikingly stable. This suggests that turnout inequality is not driven solely by specific candidates or campaign mobilization. Instead, it reflects deeper structural differences across neighborhoods.

Part 8 – Visualizations

8.1 Choropleth Maps (2024)

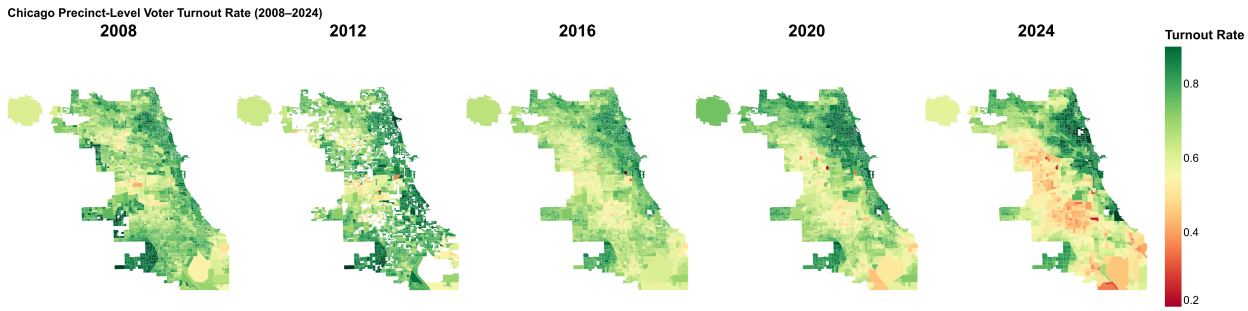

Figure 1: Avg turnout rate by precincts

Socioeconomic status (SES) and turnout

To explore whether neighborhood socioeconomic conditions are associated with voter participation, we examine the relationship between turnout and the SES index across election years The scatter plots show a strong and consistent positive relationship between socioeconomic status and turnout. Precincts with higher SES scores tend to exhibit higher turnout rates, while precincts with lower SES scores generally show lower participation.

This relationship appears in every election cycle in our dataset. While turnout increases during highly competitive elections, the relative gap between high-SES and low-SES precincts remains visible.These results suggest that socioeconomic conditions are strongly associated with participation. Neighborhood characteristics such as income levels, educational attainment, and age composition likely shape both the resources and institutional familiarity that facilitate voter engagement.

8.2 Scatter plot-Turnout vs Socioeconomic index by year

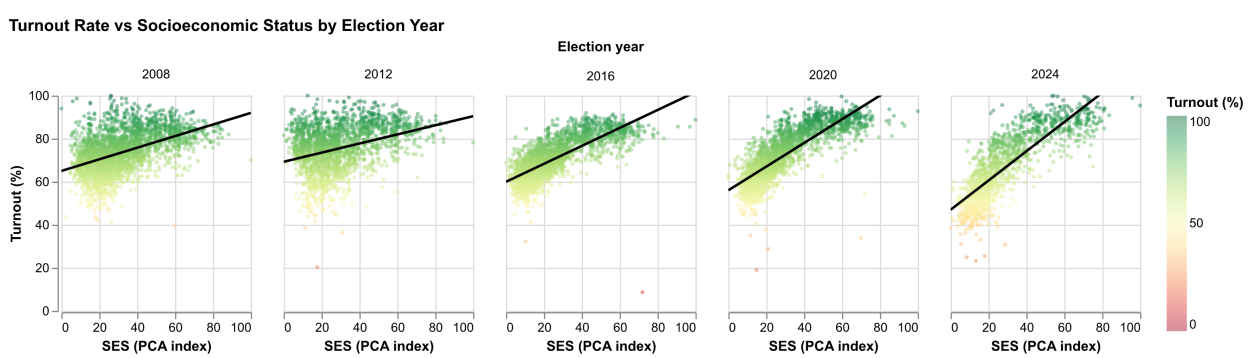

Figure 2: Avg turnout rate against socioeconomic index

Fixed Effects Regression Model

To quantify this relationship more formally, we estimate a two-way fixed effects panel regression that controls for both precinct-level and election-year differences. The regression results indicate a positive

and statistically significant relationship between socioeconomic status and turnout. Substantively, the estimates suggest that if a precinct becomes one standard deviation more socioeconomically advantaged, turnout increases by approximately 3.3 percentage points. The results reinforce the descriptive patterns observed in the visualizations.

8.3 Two-way Fixed Effects Panel Regression

Parameter Estimates

	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
const	0.7223	2.183e-17	3.309e+16	0.0000	0.7223	0.7223
ses_pca	0.0336	0.0019	17.675	0.0000	0.0298	0.0373

Demographic patterns in turnout

Finally, we examine how turnout correlates with several demographic characteristics across election cycles. The heatmap summarizes the Spearman correlation between precinct turnout rates and key demographic variables for each election year.

Several patterns emerge clearly. Income and education show the strongest and most consistent positive relationships with turnout across election cycles. Precincts with higher household incomes and larger shares of college-educated residents consistently exhibit higher participation rates. By contrast, several demographic characteristics are negatively associated with turnout. Precincts with higher shares of renters and younger residents tend to exhibit lower participation rates. Similarly, the share of Black residents shows a negative relationship with turnout in most election years, particularly in more recent elections.

Overall, the heatmap confirms that turnout differences across Chicago are strongly associated with structural neighborhood characteristics.

8.4 Heatmap: Demographics vs Turnout Rate

Policy Implications and Recommendations

These findings highlight persistent inequalities in political participation across Chicago neighborhoods. Lower-socioeconomic areas consistently exhibit lower turnout, suggesting that barriers to participation remain concentrated in particular parts of the city.

Education emerges as the most consistent predictor of turnout differences. This suggests that turnout gaps may reflect differences in civic knowledge, access to political information, and familiarity with voting institutions rather than purely financial constraints.

These patterns have several policy implications

Civic engagement efforts could be more effectively targeted toward lower-SES precincts where participation gaps are largest. Community-based outreach programs, voter information campaigns, and neighborhood organizing initiatives may help reduce informational barriers and encourage

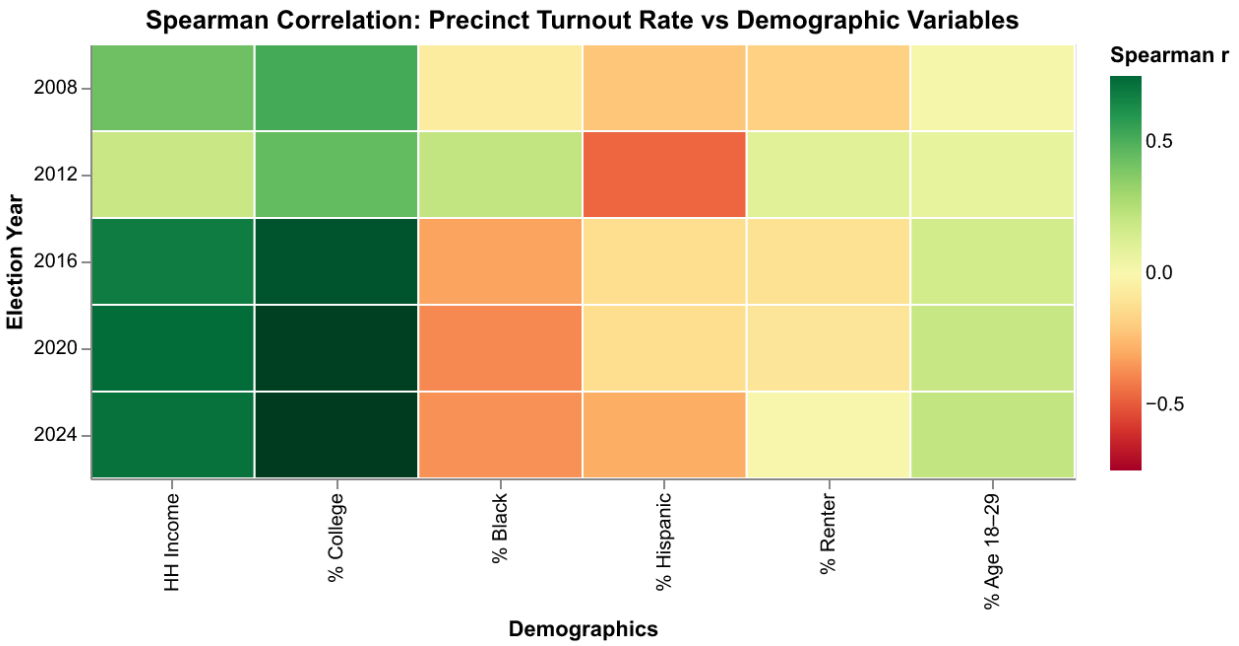

Figure 3: Spearman correlations between precinct turnout rate and demographic variables by year

participation. Besides, civic education initiatives, particularly those focused on younger voters, may help address the lower turnout observed in precincts with higher shares of young residents. Programs that improve understanding of the voting process and emphasize the importance of participation may strengthen long-term engagement.

Importantly, these policies are not solely the responsibility of government agencies. Civic organizations, campaigns, and community groups also play critical roles in mobilizing participation in targetted areaa.

Interactive Dashboard and Policy Use

In addition to the static analysis presented in this report, we developed an interactive dashboard that allows users to explore Chicago's voter turnout patterns across precincts and election years.

This tool can support election administrators, local governments, and civic organizations in identifying neighborhoods where voter engagement initiatives may have the greatest impact. It can also help campaign strategists and mobilization groups locate areas where turnout potential appears underutilized and where targeted outreach may be most effective.

By presenting these patterns in an interactive format, the dashboard translates complex data into practical insights that can inform more targeted policy interventions.

Limitations

Several limitations should be considered when interpreting these results.

- First, the analysis identifies associations rather than causal relationships. Turnout is influenced by many factors not captured in this dataset, including campaign mobilization, candidate characteristics, and historical voting behavior.

- Second, demographic characteristics are estimated using area-weighted interpolation from census tracts, which introduces some measurement uncertainty when geographic boundaries do not align perfectly.
- Finally, voter participation is shaped by complex social and institutional factors that extend beyond the scope of this analysis.

Conclusion

Despite these limitations, the analysis provides a clear picture of persistent turnout inequality across Chicago. By combining spatial analysis, statistical modeling, and interactive visualization, this project provides a foundation for understanding who votes in Chicago, and where efforts to strengthen civic participation may be most needed.